

Training-Run Sign-Up Sheet

Cross-Generator Generalization in Deepfake Detection

IE7374: Generative AI, Northeastern University

The study needs eight training runs: two models across the four held-out methods. Each run trains on the three listed methods plus all real videos and is tested on the held-out method, which it never sees during training. Put your name in the Owner column; aim for one or two runs each so the eight are shared evenly.

Run	Model	Held-out method (unseen test)	Trained on (3 methods + real)	Owner	Done
1	EfficientNet	DeepFakes	F2F, FS, NT, real		
2	EfficientNet	Face2Face	DF, FS, NT, real		
3	EfficientNet	FaceSwap	DF, F2F, NT, real		
4	EfficientNet	NeuralTextures	DF, F2F, FS, real		
5	XceptionNet	DeepFakes	F2F, FS, NT, real		
6	XceptionNet	Face2Face	DF, FS, NT, real		
7	XceptionNet	FaceSwap	DF, F2F, NT, real		
8	XceptionNet	NeuralTextures	DF, F2F, FS, real		

Legend: DF = DeepFakes, F2F = Face2Face, FS = FaceSwap, NT = NeuralTextures. Rows 1 to 4 are EfficientNet; rows 5 to 8 are XceptionNet.

Keep these identical across all eight runs

So the eight results can be compared in one transfer matrix, every run must use the same settings except for the held-out method:

Dataset and compression	FaceForensics++ at c23 (high quality)
Preprocessing and cache	Same face-crop pipeline and shared cached crops for every run
Split	Official 720 / 140 / 140 split (identity-disjoint)
Random seed	Same fixed seed, recorded in the config file
Metrics	AUC as the primary metric, reported at both frame and video level, plus a confusion matrix
Test-set discipline	The held-out method is never opened during training or tuning

Note: optional in-distribution baselines (each model trained on all four methods) can be added later for reference; they are not part of the eight runs above.